Muhammad Bilal Ali

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Education

Carnegie Mellon University — Pittsburgh, US
Masters in Privacy Engineering and AI Governance — *Aug 2025 to Dec 2026*

- Coursework: Engineering Privacy in Software, Usable Privacy and Security, Distributed Systems (Java, Apache Spark), Deep Learning, Database Management

Lahore University of Management Sciences — Lahore, Pakistan
Bachelor of Science in Computer Science; Minor in Economics — *Aug 2020 to Jul 2025*

- Coursework: Algorithms, Data Structures, Network Security, Computer Organization, Artificial Intelligence, Cloud Development
- **President, Publications Society:** Led a 20+ member editorial team, set content strategy for publications reaching the whole undergraduate body, and secured brand sponsorships for society events.
- **Teaching Assistant:** Reviewed and debugged code from 500+ students across five courses over four semesters.

Experience

Carnegie Mellon University, S3D Department — Pittsburgh, US
Research Assistant | LLM Pipelines, Privacy Threat Modeling, Python — *Jun 2026 to Aug 2026*

- Built an **LLM pipeline** that finds privacy threats in system designs and source code using LINDDUN Pro, linking each to the GDPR, CCPA, or COPPA requirement it affects. **Presented at IEEE ISoPE 2026**.
- Evaluated it against expert analysis from NIST SP 1800 on recall (missed threats), precision (false alarms), and citation correctness; used ablation studies to trace errors to the model, retrieved context, or prompts.

INDITECH MIDDLE EAST — Dubai, UAE
Data and Analytics Intern | SQL, Python, AWS, Power BI, Tableau — *Jun 2023 to Aug 2023*

- Wrote **100+ SQL queries** and built **AWS data pipelines** merging separate sales and inventory systems; traced the root causes of unreliable reports and fixed them, applying access controls and data minimization.
- Built **Power BI and Tableau dashboards** for management that led to **fewer overstock incidents**.

Projects and Research

EasyDine: Recipe Sharing App | App website — *In review: Google Play and Apple App Store*

- Designed and built a mobile app **end to end**: UI design, user accounts and authentication, recipe sharing, backend services, database design, and the website; currently submitted for both app stores for review.
- Integrated an **LLM agent** which can assist the user in use any features efficiently, such as planning meals and tracking calories.

Privacy Preference Tool for Smart Home Devices | Github Link — *CMU Capstone with CableLabs*

- Co-developing a prototype of the proposed IETF Privacy Preference Declaration framework so households can control what smart devices do with their data: a web tool, a preference vocabulary, and a JSON Schema validator.

Membership Inference Attack on a Fine-Tuned LLM | Github Link — *PyTorch, Hugging Face, PEFT*

- Attacked a **GPT-2 model** fine-tuned with **LoRA** to reveal its training data; trained 4 shadow models matched to the target, scored by per-token loss and perplexity.

Second Brain: RAG and AI Agent Tooling | Github Link — *Python, FastAPI, Qdrant, MCP*

- Built a RAG service that searches PDFs, Notion, YouTube, and web pages in one query, combining **Qdrant** vector search, BM25, and rank fusion with **cross-encoder** re-ranking for more relevant results.
- Connected it to Claude through a custom **MCP server** with multi-query expansion, serving streaming answers via FastAPI that cite the source behind every claim.

Skills

Languages and Frameworks: Python, Java, C/C++, Go, SQL, JavaScript, React, Node.js, FastAPI, PyTorch, Hugging Face, .NET
Privacy and Security: Threat modeling (LINDDUN Pro), ISO 27701, GDPR, CCPA, COPPA
AI and Data: LLM pipelines, AI agents (MCP), RAG, LLM evaluation, Apache Spark, Power BI, Tableau
Infrastructure and Tools: PostgreSQL, MongoDB, Qdrant, Docker, AWS, Git, Linux